Implementation of Indoor Positioning System

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***Abstract*—This documentation presents the development of a real-time indoor positioning system (IPS) utilizing Wi-Fi Fine Time Measurement (FTM) technology on ESP32 microcontrollers. Because traditional global positioning systems fail to provide reliable coverage in indoor environments, this project leverages Time-of-Flight (ToF) ranging techniques to calculate distances between a mobile ESP32 tag and a network of stationary Wi-Fi anchors. Due to severe indoor multipath interference caused by 2.4 GHz signal reflections, traditional geometric trilateration and linear regression models were ultimately superseded by a purely data-driven K-Nearest Neighbors (KNN) RF Fingerprinting machine learning approach. By mapping ambient multipath signatures, the KNN algorithm dynamically computes the mobile tag's X and Y spatial coordinates, achieving stable, sub-meter localization. The positioning logic is edge-computed by the integrated microcontroller, which concurrently operates as a local web server to host a live HTML5 canvas interface for real-time visualization. Ultimately, this system demonstrates a robust and low-latency solution for continuous indoor tracking, successfully overcoming the geometric limitations of narrow-band Wi-Fi signals.**

***Index Terms*—Ultra-Wideband (UWB), Indoor Positioning System (IPS), Time-of-Flight (ToF), Trilateration Algorithm, Machine Learning, K-Nearest Neighbors (KNN), Real-Time Spatial Tracking, Edge Computing**

1 Introduction

1.1 Background and Motivation

The proliferation of location-based services has driven a massive demand for accurate spatial tracking. While the Global Positioning System (GPS) is highly effective for outdoor navigation, it is fundamentally flawed in indoor environments. Satellite signals suffer from severe attenuation when passing through building materials, and the presence of walls and furniture creates complex multipath propagation. Consequently, GPS and other traditional radio-frequency tracking methods often yield positional errors measured in meters, which is unacceptable for precise indoor applications such as autonomous navigation, asset tracking, and smart environment automation. This creates a critical need for an alternative localization technology designed specifically for the unique challenges of indoor spaces.

1.2 Ultra-Wideband (UWB) Technology

To bridge the gap in indoor localization, Ultra-Wideband (UWB) technology has emerged as a superior solution. Unlike traditional narrowband systems that rely on signal strength (RSSI) to estimate distance, UWB operates across a massive frequency spectrum (typically 500 MHz or more) using extremely short, nanosecond-scale radio pulses. This unique physical characteristic allows UWB receivers to accurately distinguish between the direct line-of-sight signal and delayed multipath reflections. By measuring the precise Time-of-Flight (ToF) of these pulses traveling at the speed of light, UWB systems can calculate physical distances with centimeter-level accuracy, making it an ideal foundation for high-precision tracking.

1.3 Project Scope and Objectives

This project details the design and implementation of a custom UWB-based Indoor Positioning System (IPS). The primary objective is to develop a reliable, edge-computed tracking solution capable of real-time spatial localization. The system architecture consists of a mobile UWB tag and a network of fixed UWB anchors. By continually polling the ToF between the tag and the anchors, the system generates high-fidelity distance measurements. These ranges are actively processed through positioning algorithms on a central microcontroller to compute the exact Cartesian coordinates (X and Y) of the target. Finally, to ensure accessibility and immediate visual feedback, the hardware concurrently operates as a local web server, broadcasting the computed coordinates to a live HTML5 graphical interface.

2 Methodology

2.1 System Architecture

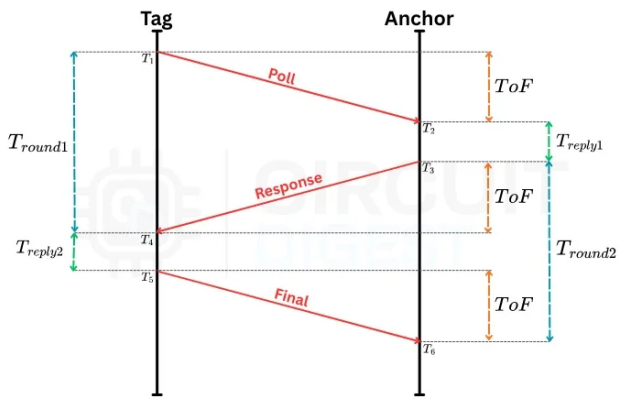
The proposed Indoor Positioning System (IPS) is built upon a centralized edge-computing architecture. The physical setup consists of three stationary Ultra-Wideband (UWB) reference nodes, designated as anchors, and one mobile UWB node, designated as the tag. The tag continuously broadcasts ranging pulses to the anchors, which process the signals to determine precise radial distances. An integrated microcontroller (acting as the central processing unit) aggregates these raw distance metrics, applies software-level outlier rejection, and computes the final spatial coordinates.

2.2 Time-of-Flight (ToF) Ranging

UWB technology relies on the transmission of nanosecond-scale radio pulses. By measuring the exact time it takes for a pulse to travel from the tag to an anchor and back, the system calculates the physical distance r. This relationship is fundamentally defined by the equation:

$$r = c \times \frac{t_{ToF}}{2} = c \times \frac{(T_{recv} - T_{sent}) - \text{Delay}}{2} \tag{1}$$

where c is the speed of light in a vacuum ($\approx 3 \times 10^8$ m/s), and t_{ToF} is the total round-trip time.



2.3 2D Trilateration Algorithm

Once the radial distances from the three anchors (r_1, r_2, r_3) are acquired, an initial 2D trilateration algorithm was employed to find the tag's exact spatial intersection. To optimize the computational load on the microcontroller, the coordinate system is aligned such that Anchor 1 is at the origin $(0, 0)$, Anchor 2 lies on the x-axis at $(d, 0)$, and Anchor 3 is located at an arbitrary point (i, j).

The system of geometric equations representing the three intersecting ranging circles is:

$$r_1^2 = x^2 + y^2 \tag{2}$$

$$r_2^2 = (x - d)^2 + y^2 \tag{3}$$

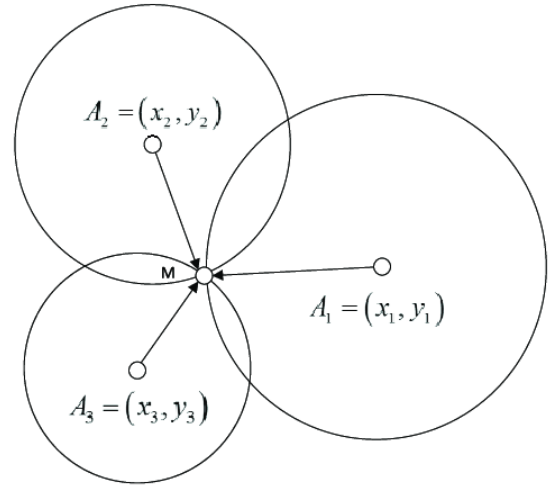
$$r_3^2 = (x - i)^2 + (y - j)^2 \tag{4}$$

By expanding and algebraically manipulating these equations, the exact x and y coordinates of the mobile tag can be computed dynamically. The microcontroller executes the following simplified formulas in real-time:

$$x = \frac{r_1^2 - r_2^2 + d^2}{2d} \tag{5}$$

$$y = \frac{r_1^2 - r_3^2 + i^2 + j^2 - 2ix}{2j} \tag{6}$$

This streamlined algebraic approach provides a highly deterministic and computationally efficient method for real-time edge processing without the need for complex matrix inversions.



3 IMPLEMENTATION USING SDR

To gain full control over the physical layer, dedicated UWB modules were replaced with ADALM-Pluto Software Defined Radios (SDRs). The SDRs were unlocked via SSH to support frequencies up to 6 GHz , and all signal processing logic was transitioned to Python using `numpy`.

For Time-of-Flight ranging, the system utilizes a custom Zadoff-Chu pulse. The Tag transmits this sequence while initiating a timer, and the Anchor immediately reflects it back. Upon reception, the Tag performs a digital cross-correlation against the known sequence to pinpoint the exact peak arrival time, thereby determining the signal's round-trip time.

3.1 Critical Challenges and Observations

Transitioning from hardware-timed UWB modules to Software Defined Radios introduced several severe bottlenecks and processing latency issues:

- **USB Power Bottleneck:** Two ADALM-Pluto SDRs draw approximately 1A of combined current, which caused standard laptop ports to brown out and repeatedly crash the network. This was resolved by connecting the devices to entirely separate USB buses on different sides of the host machine.
- **Network Routing Conflicts:** The Ubuntu host OS struggled to route traffic simultaneously to two different subnets (2.x and 3.x) on a single interface. Manual IP aliasing using the `sudo ip addr add` command was required to force the connections.
- **The "100 km" Offset Error:** Initial testing consistently reported massive distance measurements between 100 km and 200 km, despite the SDRs being placed exactly 1 meter apart. This fundamental error was primarily driven by system latency. USB data transfers and Python processing added an approximate 0.33 ms delay. Because radio waves travel at 300 km/ms, this microsecond delay translated mathematically into a $\approx$ 100 km physical offset.

- **False Noise Detection and CFO:** Without a defined correlation threshold, the receiver triggered on random RF noise spikes instead of the true signal, returning erratic distances. Furthermore, unsynchronized clocks between the Tag and Anchor caused Carrier Frequency Offset (CFO), which smeared the Zadoff-Chu sequence's correlation peak and caused the algorithm to miss the true arrival time.
- **Processing Jitter:** Because standard Linux distributions are not Real-Time Operating Systems (RTOS), background tasks and OS scheduling paused the Python script randomly. Coupled with USB 2.0 polling latency ($\pm 125\mu s$), the processing time was highly variable. A timing jitter of just $1\mu s$ creates a 300-meter distance error, demonstrating the extreme instability of software-reliant Time-of-Flight ranging.

3.2 Results

In conclusion, replacing dedicated hardware with Software Defined Radios (SDRs) like the ADALM-Pluto fundamentally fails for Time-of-Flight (ToF) ranging due to insurmountable system latencies. Because electromagnetic waves travel at light speed, the inherent 0.33 ms delay from USB transfers and Python processing mathematically fabricates massive distance errors of roughly 100 kilometers. Furthermore, relying on a non-Real-Time Operating System (RTOS) introduces unpredictable processing jitter, while unsynchronized clocks cause Carrier Frequency Offsets (CFO) that smear the correlation peaks and trigger false noise detections. Ultimately, while SDRs offer excellent physical layer control, their tethered, software-reliant architecture is simply too unstable to support the microsecond-level precision required for functional UWB positioning.

4 IMPLEMENTATION USING ESP32: ANALYTICAL APPROACH

4.1 HARDWARE REQUIREMENTS

- **Tracking Tag:** ESP32 Microcontroller (FTM/802.11mc compatible, e.g., ESP32-S2 or ESP32-S3).
- **Anchors:** 3x Wi-Fi Access Points or ESP32 nodes configured as FTM responders.
- **Monitoring Device:** Smartphone or PC with Wi-Fi capabilities and a modern web browser.

4.2 SYSTEM ARCHITECTURE

The positioning system relies on calculating the distance between the mobile tag and three fixed anchors located at known coordinates: A1(0,0), A2(10,0), and A3(5,8).

4.2.1 Time of Flight (ToF) Measurement: The ESP32 initiates an FTM burst (typically 32 packets) to each anchor. By measuring the Round Trip Time (RTT) of these radio frequency bursts, the distance is mathematically derived using the speed of light.

4.2.2 Asynchronous Web Server: To prevent blocking operations during FTM packet collection, the ESP32 hosts a local SoftAP network (`ESP32-Live-Map`). A lightweight web server handles incoming client requests and serves a dynamic HTML5 UI that fetches JSON coordinate data.

4.3 FILTERING AND GEOMETRIC TRILATERATION

4.3.1 Moving Median Filter: To isolate the true line-of-sight distance from anomalous readings, a 5-point rolling window is utilized. The array is sorted using a bubble sort algorithm, and the exact median value is extracted to strip sudden multipath spikes.

4.3.2 2D Trilateration Math: The intersecting X and Y coordinates of the tag are calculated using standard algebraic trilateration derivations based on the filtered radii:

$$x = \frac{r_1^2 - r_2^2 + d^2}{2d}$$

$$y = \frac{r_1^2 - r_3^2 + i^2 + j^2 - 2ix}{2j}$$

where d is the x-coordinate of Anchor 2, and (i, j) are the coordinates of Anchor 3.

4.4 ANCHOR NODE CODE

Each stationary anchor runs the following firmware. On startup, the anchor initialises ESP-NOW and registers a receive callback. Whenever a ranging packet arrives from the tag, the anchor automatically registers the sender as a peer (if not already known) and immediately reflects the exact same data structure back, enabling the tag to compute the round-trip time.

```
#include <esp_now.h>
#include <WiFi.h>

// Structure to hold incoming data.
// CRITICAL: This exact structure must also be used on your main sending device!
typedef struct struct_message {
  int id;
  float x;
  float y;
} struct_message;

struct_message myData;

// Callback function executed when data is received
void OnDataRecv(const uint8_t * mac, const uint8_t * incomingData, int len) {
  // 1. Copy the incoming bytes into our data structure
  memcpy(&myData, incomingData, sizeof(myData));

  Serial.print("Received packet from MAC: ");
  for (int i = 0; i < 6; i++) {
    Serial.print(mac[i], HEX);
    if (i < 5) Serial.print(":");
  }
  Serial.println();

  // 2. Check if the sender is already registered as a peer.
  //    If not, add them on the fly.
  if (!esp_now_is_peer_exist(mac)) {
    esp_now_peer_info_t peerInfo;
    memset(&peerInfo, 0, sizeof(peerInfo));
    memcpy(peerInfo.peer_addr, mac, 6);
    peerInfo.channel = 0;  // Use current Wi-Fi channel
    peerInfo.encrypt = false;

    if (esp_now_add_peer(&peerInfo) != ESP_OK) {
      Serial.println("Failed to add peer for reflection.");
      return;
    }
  }

  // 3. Reflect the exact same data back to the sender
  esp_err_t result = esp_now_send(mac, (uint8_t *) &myData, sizeof(myData));

  if (result == ESP_OK) {
    Serial.println("Packet successfully reflected.");
  } else {
    Serial.println("Error bouncing packet back.");
  }
}

void setup() {
  Serial.begin(115200);

  // Set ESP32 to Wi-Fi Station mode
  WiFi.mode(WIFI_STA);

  // Print the MAC Address of this Anchor so you know what to target
  Serial.print("ANCHOR MAC ADDRESS: ");
  Serial.println(WiFi.macAddress());

  // Initialize ESP-NOW
  if (esp_now_init() != ESP_OK) {
    Serial.println("Error initializing ESP-NOW");
    return;
  }

  // Register the receive callback function
  esp_now_register_recv_cb(OnDataRecv);
}

void loop() {
  // The loop does nothing. The anchor sits quietly and waits
  // for the hardware interrupt (the callback) to trigger when
  // a packet arrives.
  delay(1000);
}
```

4.5 CODE IMPLEMENTED

```
/* --- Analytical approach for UWB Positioning system ---
   Anchors: A1(0,0), A2(10,0), A3(5,8)

*/

#include "WiFi.h"
#include <WebServer.h>
#include <math.h>

// ==========================================
// 1. CONFIGURATION
// ==========================================
float anchors[3][2] = {
  {0.00,  0.00},  // Anchor 01
  {10.00, 0.00},  // Anchor 02
  {5.00,  8.00}   // Anchor 03
};
String anchorNames[] = {"FTM_Anchor_01", "FTM_Anchor_02", "FTM_Anchor_03"};
float distance_offset = 0.0;

// ==========================================
// 2. FILTERING & CACHED VARIABLES
// ==========================================
float dist_history[3][5];
int history_idx[3] = {0, 0, 0};
float smoothed_distances[3] = {0, 0, 0};
bool first_reading[3] = {true, true, true};

volatile bool got_data = false;
float last_raw_val = 0;

// Cached Wi-Fi Data (Prevents scanning in the loop)
uint8_t anchorBSSIDs[3][6];
int anchorChannels[3];
bool anchorFound[3] = {false, false, false};

// Coordinate Globals for Web UI
float posX = 0.0;
float posY = 0.0;

WebServer server(80);

// ==========================================
// 3. THE WEB UI (HTML + JAVASCRIPT)
// ==========================================
const char index_html[] PROGMEM = R"rawliteral(
<!DOCTYPE html>
<html>
<head>
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>ESP32 Indoor GPS</title>
  <style>
    body { font-family: Arial; text-align: center; background-color: #222; color: #fff; margin: 0; padding: 20px;}
    canvas { background-color: #111; border: 2px solid #555; border-radius: 5px; box-shadow: 0px 0px 15px rgba(0,255,255,0.2); margin-top: 10px; max-width: 100%;}
    .stats { font-size: 1.2em; font-weight: bold; color: #00FFCC; margin-top: 15px; }
    .debug { font-size: 0.9em; color: #aaa; margin-top: 5px; }
  </style>
</head>
<body>
  <h2>Live Tracking (10m x 8m)</h2>

  <canvas id="mapCanvas" width="400" height="320"></canvas>

  <div class="stats" id="coords">X: 0.00 m | Y: 0.00 m</div>
  <div class="debug" id="dists">A1: 0m | A2: 0m | A3: 0m</div>

  <script>
    const canvas = document.getElementById('mapCanvas');
    const ctx = canvas.getContext('2d');
    const scale = 40; // 40 pixels = 1 meter

    function drawGrid() {
      ctx.clearRect(0, 0, canvas.width, canvas.height);

      // Draw Grid Lines
      ctx.strokeStyle = "#333";
      for(let i=0; i<10; i++) {
        ctx.beginPath(); ctx.moveTo(i*scale, 0); ctx.lineTo(i*scale, 320); ctx.stroke();
        ctx.beginPath(); ctx.moveTo(0, i*scale); ctx.lineTo(400, i*scale); ctx.stroke();
      }

      // Draw Anchors
      ctx.fillStyle = "#00FFCC";
      ctx.fillRect(0, 320 - 10, 10, 10); // A1 (0,0)
      ctx.fillRect(10 * scale - 10, 320 - 10, 10, 10); // A2 (10,0)
      ctx.fillRect(5 * scale - 5, 320 - (8 * scale), 10, 10); // A3 (5,8)

      ctx.fillStyle = "#fff";
      ctx.font = "12px Arial";
      ctx.fillText("A1(0,0)", 15, 310);
      ctx.fillText("A2(10,0)", 340, 310);
      ctx.fillText("A3(5,8)", 215, 20);
    }

    function updateMap() {
      fetch('/data')
        .then(response => response.json())
        .then(data => {
          drawGrid();

          // Draw Tag
          let cx = data.x * scale;
          let cy = 320 - (data.y * scale); // Invert Y for canvas

          // Constrain dot to canvas bounds visually
          if(cx < 0) cx = 5; if(cx > 400) cx = 395;
          if(cy < 0) cy = 5; if(cy > 320) cy = 315;

          ctx.beginPath();
          ctx.arc(cx, cy, 8, 0, 2 * Math.PI);
          ctx.fillStyle = "#FF3366";
          ctx.fill();
          ctx.strokeStyle = "#fff";
          ctx.stroke();

          document.getElementById('coords').innerText = "X: " + data.x.toFixed(2) + " m | Y: " + data.y.toFixed(2) + " m";
          document.getElementById('dists').innerText = "A1: " + data.r1.toFixed(1) + "m | A2: " + data.r2.toFixed(1) + "m | A3: " + data.r3.toFixed(1) + "m";
        })
        .catch(err => console.log(err));
    }
```

```
    drawGrid();
    setInterval(updateMap, 400); // Fetch data every 0.4s
  </script>
</body>
</html>
)rawliteral";

// ==========================================
// 4. FTM EVENT HANDLER
// ==========================================
void onFtmReport(arduino_event_t *event) {
  wifi_event_ftm_report_t *report = &event->event_info.wifi_ftm_report;
  if (report->status == FTM_STATUS_SUCCESS) {
    last_raw_val = (report->dist_est / 100.0) + distance_offset;
    got_data = true;
  }
}

// ==========================================
// 5. SETUP
// ==========================================
void setup() {
  Serial.begin(115200);

  // Wi-Fi Setup: AP mode for Web Server, STA mode for FTM
  WiFi.mode(WIFI_AP_STA);
  WiFi.softAP("ESP32-Live-Map", "12345678"); // Hotspot Name & Password
  WiFi.onEvent(onFtmReport, ARDUINO_EVENT_WIFI_FTM_REPORT);
  WiFi.setSleep(false);

  // Web Server Endpoints
  server.on("/", []() {
    server.send(200, "text/html", index_html);
  });

  server.on("/data", []() {
    char json[100];
    sprintf(json, "{\"x\":%.2f, \"y\":%.2f, \"r1\":%.2f, \"r2\":%.2f, \"r3\":%.2f}",
            posX, posY, smoothed_distances[0], smoothed_distances[1], smoothed_distances[2]);
    server.send(200, "application/json", json);
  });

  server.begin();

  Serial.println("\n--- 10m Stabilized System Starting ---");
  Serial.println("Scanning for Anchors...");

  // ONE-TIME SCAN FOR ANCHORS
  int n = WiFi.scanNetworks(false, false, false, 150);
  for (int i = 0; i < n; i++) {
    for (int j = 0; j < 3; j++) {
      if (WiFi.SSID(i) == anchorNames[j]) {
        anchorChannels[j] = WiFi.channel(i);
        memcpy(anchorBSSIDs[j], WiFi.BSSID(i), 6);
        anchorFound[j] = true;
        Serial.printf("Found %s on Channel %d\n", anchorNames[j].c_str(), anchorChannels[j]);
      }
    }
  }
  WiFi.scanDelete(); // Free memory

  Serial.println("Connect Phone to Wi-Fi: ESP32-Live-Map");
  Serial.println("Open Browser to: http://192.168.4.1");
}

// ==========================================
// 6. MAIN LOOP
// ==========================================
void loop() {
  server.handleClient(); // Keep the website alive

  for (int j = 0; j < 3; j++) {
    if (anchorFound[j]) {
      got_data = false;

      // Request FTM using CACHED BSSID and Channel
      if (WiFi.initiateFTM(32, 0, anchorChannels[j], anchorBSSIDs[j])) {
        unsigned long start = millis();

        // Wait for data, feeding the web server
        while(!got_data && millis() - start < 1000) {
          delay(10);
          server.handleClient();
        }

        if(got_data && last_raw_val > 0.1 && last_raw_val < 30.0) {

          // Fill array on first reading to prevent ramp-up lag
          if (first_reading[j]) {
            for (int k = 0; k < 5; k++) dist_history[j][k] = last_raw_val;
            first_reading[j] = false;
          } else {
            dist_history[j][history_idx[j]] = last_raw_val;
            history_idx[j] = (history_idx[j] + 1) % 5;
          }

          // --- MEDIAN FILTER LOGIC ---
          float temp[5];

          // 1. Copy the history array
          for(int k=0; k<5; k++) {
            temp[k] = dist_history[j][k];
          }

          // 2. Bubble Sort the temporary array
          for(int p=0; p<4; p++) {
            for(int q=0; q<4-p; q++) {
              if(temp[q] > temp[q+1]) {
                float t = temp[q];
                temp[q] = temp[q+1];
                temp[q+1] = t;
              }
            }
          }

          // 3. Extract the median value (middle element)
          smoothed_distances[j] = temp[2];
          // --------------------------
        }
      }

      // Brief delay, feeding server
      unsigned long d_start = millis();
      while(millis() - d_start < 50) { server.handleClient(); delay(1); }
    }
  }
```

```

  // Only calculate XY if we have valid readings for all 3 anchors
  if (!first_reading[0] && !first_reading[1] && !first_reading[2]) {
    calculateXY();
  }
}

// ========================================
// 7. TRILATERATION MATH
// ========================================
void calculateXY() {
  float r1 = smoothed_distances[0];
  float r2 = smoothed_distances[1];
  float r3 = smoothed_distances[2];

  float d = anchors[1][0]; // 10.0
  float i = anchors[2][0]; // 5.0
  float j = anchors[2][1]; // 8.0

  float x = (pow(r1, 2) - pow(r2, 2) + pow(d, 2)) / (2 * d);
  float y = (pow(r1, 2) - pow(r3, 2) + pow(i, 2) + pow(j, 2) - (2 * i * x)) / (2 * j);

  if (isnan(x) || isnan(y)) return;

  // Update globals for the Web UI
  posX = x;
  posY = y;

  Serial.printf("DIST -> A1:%.2fm A2:%.2fm A3:%.2fm | POS -> X:%.2fm, Y:%.2fm\n", r1, r2, r3, x, y);
}
```

4.6 EXPERIMENTAL RESULTS AND LIMITATIONS

While the software architecture successfully processes FTM data, empirical testing revealed severe hardware and environmental limitations that restrict the practical application of purely geometric trilateration for indoor positioning.

4.6.1 Outdoor vs. Indoor Performance: Field experiments were conducted in both clear outdoor environments and enclosed indoor spaces.

- **Outdoor Environment:** In open-air conditions with a clear Line-of-Sight (LOS) between the tag and anchors, the system demonstrated an average positioning error of 1 to 2 meters. This establishes the baseline capability of the ESP32's FTM implementation under optimal conditions.
- **Indoor Environment:** When the identical setup was transitioned indoors, the system exhibited significantly higher error rates, often reporting distances that were mathematically impossible given the physical dimensions of the room.

4.6.2 Multipath Interference and Signal Reflection: The primary cause of the severe indoor inaccuracy is multipath propagation. In an indoor environment, 2.4 GHz Wi-Fi signals reflect heavily off walls, floors, ceilings, and metallic objects. Consequently, the ESP32 often receives delayed, reflected signals rather than the direct LOS signal. Because FTM relies strictly on the Time-of-Flight, these delayed reflections artificially inflate the calculated distance, completely corrupting the trilateration geometry.

4.6.3 Impossibility of Centimeter-Level Accuracy: It must be explicitly stated that achieving centimeter-level accuracy using the ESP32 is a physical and mathematical impossibility. The resolution of FTM is directly constrained by the bandwidth of the Wi-Fi signal. Standard ESP32 hardware operates on a 20 MHz channel bandwidth, which limits the theoretical best-case distance resolution to the meter level. Unlike true Ultra-Wideband (UWB) technologies, which use wide bandwidths (500 MHz or more) to achieve very sharp, centimeter-accurate pulses, the ESP32 is fundamentally hardware-locked to meter-level accuracy.

5 IMPLEMENTATION USING ESP32: MACHINE LEARNING APPROACH

5.1 OVERCOMING GEOMETRIC LIMITATIONS

Because pure mathematical trilateration fails in multipath-heavy indoor environments, the system architecture can be adapted to utilize data-driven Machine Learning (ML) techniques. By treating the hardware biases and RF reflections as learnable patterns rather than geometric absolute values, the ESP32's meter-level limitations can be heavily mitigated.

5.2 Baseline ML: Linear Regression (Least Squares Calibration)

Similar to calibrating analog hardware sensors to account for static environmental variables, a foundational linear regression model was deployed to correct antenna phase center shifts and hardware delays.

Using the Least Squares method, the system was trained on a dataset of known physical coordinates to mathematically map the erroneous measured FTM distances to the true physical distances. The data collected for this training phase is detailed in Table I.

TABLE I: Raw FTM Distance Dataset for ML Calibration

S.NO	X	Y	Dist_A1	Dist_A2	Dist_A3
1	0	0	0.1	15.9	22.4
2	0	2	8.6	12.4	12.9
3	0	4	11.8	13.3	18.1
4	0	6	16.5	14.3	10.2
5	0	8	9.3	14.4	7.5
6	2	2	5.4	18.8	10.5
7	2	4	4.7	10.8	14.4
8	2	6	6.8	12.6	9.8
9	4	2	16.1	11.6	16.9
10	4	4	6.6	10.1	10.9
11	4	6	7.8	15.6	15.3
12	6	2	7.5	10.1	9.0
13	6	4	7.3	7.7	8.6
14	6	6	9.0	8.3	4.3
15	8	2	9.2	4.3	9.8
16	8	4	12.4	10.3	5.7
17	8	6	10.8	12.2	6.9
18	8	8	13.8	15.6	4.3
19	10	0	14.3	1.1	17.3
20	10	2	8.8	5.8	9.4
21	10	4	14.7	6.5	8.0
22	10	6	18.3	8.8	7.5
23	10	8	15.2	12.2	7.0
24	5	0	5.4	7.8	9.6
25	5	8	16.5	16.2	1.2

To derive the optimal calibration constants, the Ordinary Least Squares (OLS) method was formulated using matrix algebra. For each anchor located at (X_A, Y_A), the true Euclidean distance vector Y (25×1) for the $n = 25$ calibration points is computed as:

$$y_i = \sqrt{(X_i - X_A)^2 + (Y_i - Y_A)^2}$$

The measured FTM distances populate the design matrix X (25×2), where the first column consists of ones to account for the intercept c, and the second column contains the raw distance readings $d_{measured}$:

$$\vec{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_{25} \end{bmatrix}, \quad \vec{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{25} \end{bmatrix}, \quad \vec{\beta} = \begin{bmatrix} c \\ m \end{bmatrix}$$

The coefficient vector $\vec{\beta}$ is solved by minimizing the residual sum of squares using the closed-form normal equation:

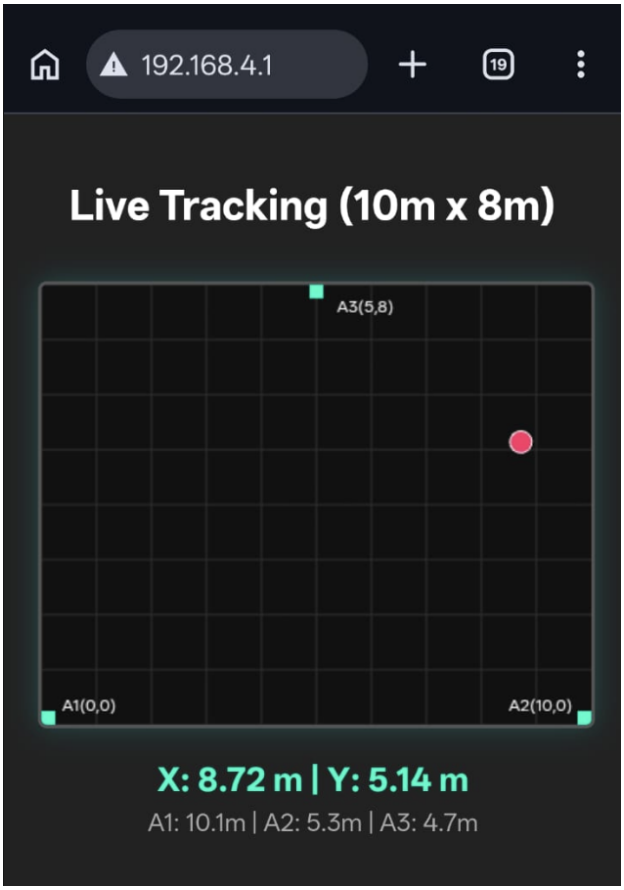
$$\vec{\beta} = (\vec{X}^T \vec{X})^{-1} \vec{X}^T \vec{Y}$$

By applying this matrix transformation to the dataset in Table I for each respective anchor—$A_1(0, 0)$, $A_2(10, 0)$, and $A_3(5, 8)$—the derived regression constants optimized for the three ESP32 anchors are:

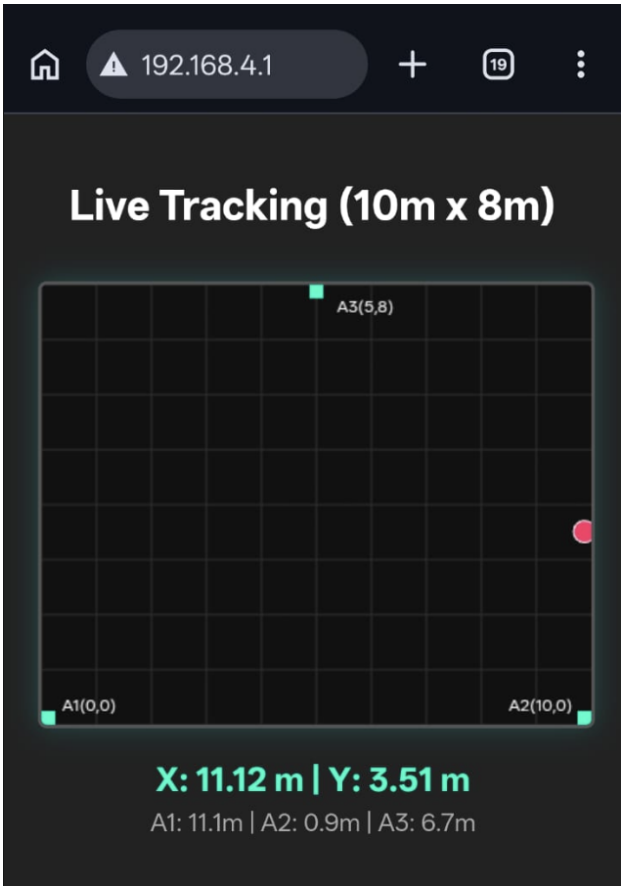
- **Anchor 1:** $m = 0.4614, c = 2.5134$
- **Anchor 2:** $m = 0.6145, c = 0.3300$
- **Anchor 3:** $m = 0.2887, c = 2.4097$

By applying these trained weights ($d_{true} = m \cdot d_{measured} + c$) before executing the trilateration algorithm, the system effectively normalizes the non-linear hardware delays that the geometric approach ignores.

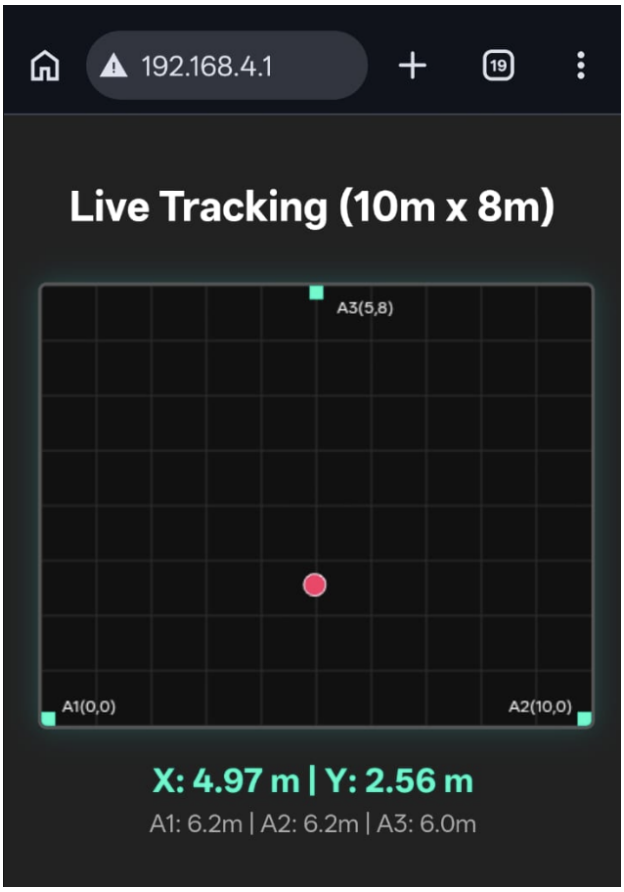
5.3 System Visualizations and Images


192.168.4.1
19
Live Tracking (10m x 8m)
A3(5,8)
A1(0,0)
A2(10,0)
X: 8.72 m | Y: 5.14 m
A1: 10.1m | A2: 5.3m | A3: 4.7m

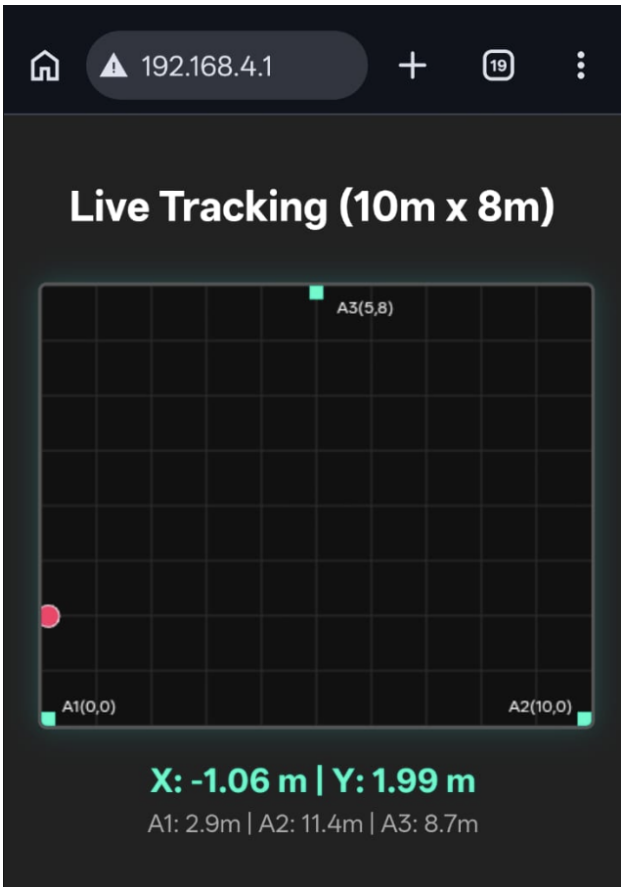
(8,6)


192.168.4.1
19
Live Tracking (10m x 8m)
A3(5,8)
A1(0,0)
A2(10,0)
X: 11.12 m | Y: 3.51 m
A1: 11.1m | A2: 0.9m | A3: 6.7m

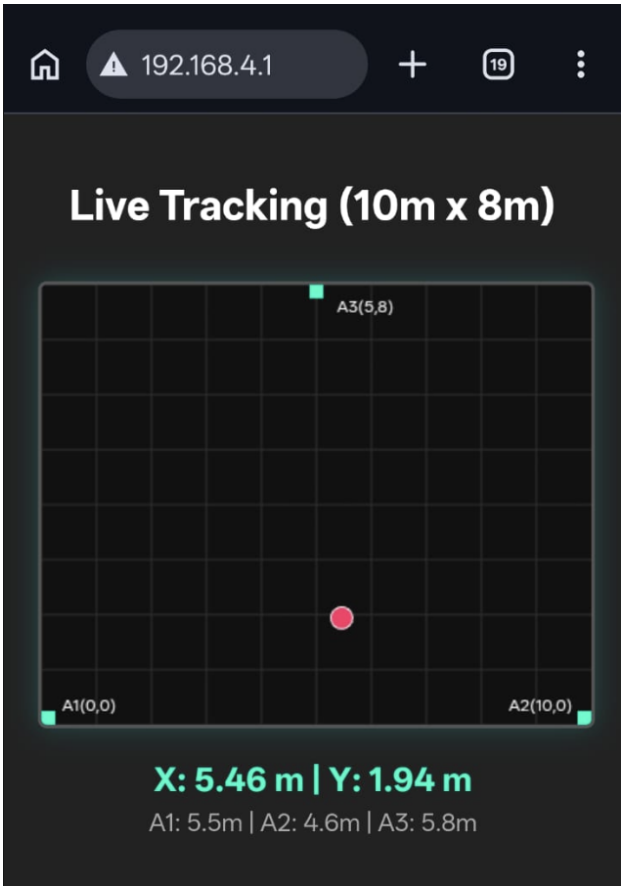
(10,0)


192.168.4.1
19
Live Tracking (10m x 8m)
A3(5,8)
A1(0,0)
A2(10,0)
X: 4.97 m | Y: 2.56 m
A1: 6.2m | A2: 6.2m | A3: 6.0m

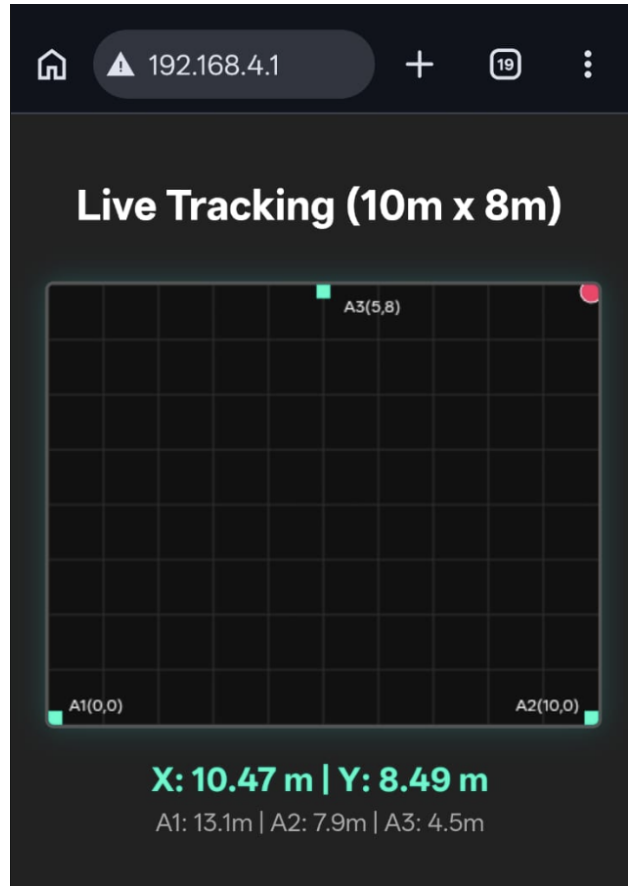
(4,4)


192.168.4.1
19
Live Tracking (10m x 8m)
A3(5,8)
A1(0,0)
A2(10,0)
X: -1.06 m | Y: 1.99 m
A1: 2.9m | A2: 11.4m | A3: 8.7m

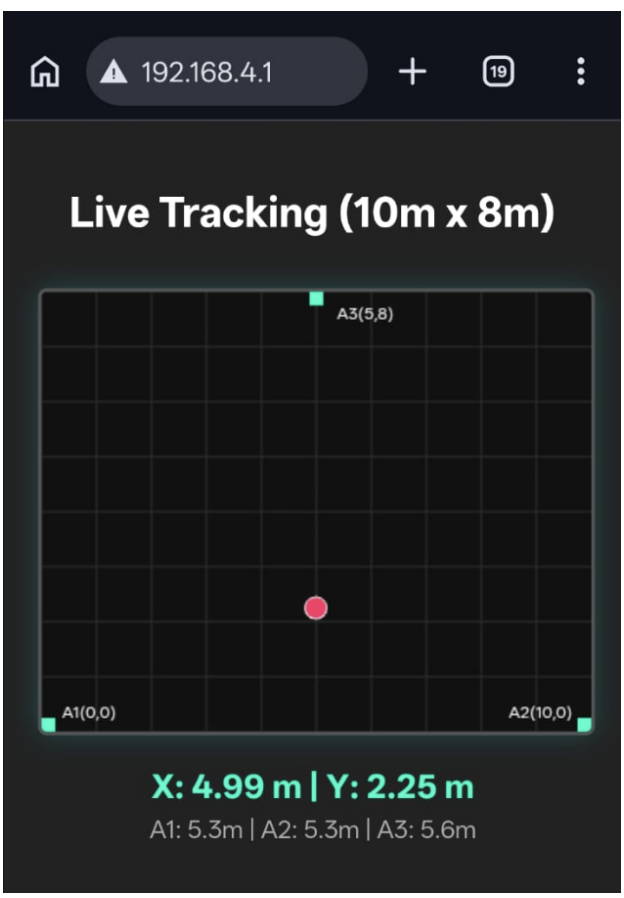
(0,0)


192.168.4.1
19
Live Tracking (10m x 8m)
A3(5,8)
A1(0,0)
A2(10,0)
X: 5.46 m | Y: 1.94 m
A1: 5.5m | A2: 4.6m | A3: 5.8m

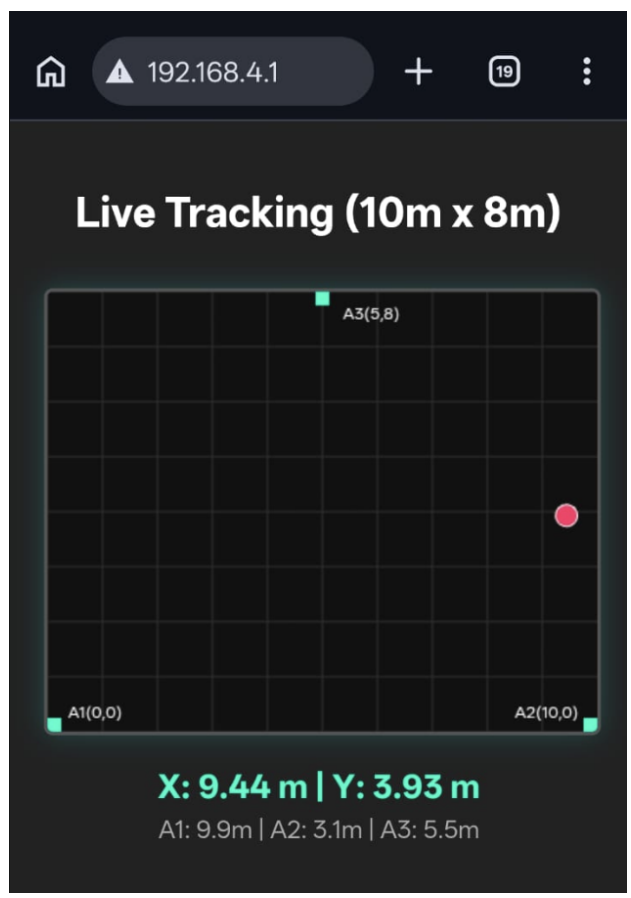
(6,2)


192.168.4.1
19
Live Tracking (10m x 8m)
A3(5,8)
A1(0,0)
A2(10,0)
X: 10.47 m | Y: 8.49 m
A1: 13.1m | A2: 7.9m | A3: 4.5m

(8,6)


192.168.4.1
19
Live Tracking (10m x 8m)
A3(5,8)
A1(0,0)
A2(10,0)
X: 4.99 m | Y: 2.25 m
A1: 5.3m | A2: 5.3m | A3: 5.6m

(4,2)


192.168.4.1
19
Live Tracking (10m x 8m)
A3(5,8)
A1(0,0)
A2(10,0)
X: 9.44 m | Y: 3.93 m
A1: 9.9m | A2: 3.1m | A3: 5.5m

(8,2)

5.4 Code Implemented

```
/* --- ML approach for UWB Positioning system ---
   Anchors: A1(0,0), A2(10,0), A3(5,8)
*/

#include "WiFi.h"
#include <WebServer.h>
#include <math.h>

// ==========================================
// 1. CONFIGURATION & CALIBRATION
// ==========================================
float anchors[3][2] = {
  {0.00,  0.00},  // Anchor 01
  {10.00, 0.00},  // Anchor 02
  {5.00,  8.00}   // Anchor 03
};
String anchorNames[] = {"FTM_Anchor_01", "FTM_Anchor_02", "FTM_Anchor_03"};

// LEAST SQUARES CALIBRATION CONSTANTS (Calculated from your 25-point dataset)
// Format: d_true = (m * d_measured) + c
float A1_m = 0.4614; float A1_c = 2.5134;
float A2_m = 0.6145; float A2_c = 0.3300;
float A3_m = 0.2887; float A3_c = 2.4097;

// ==========================================
// 2. FILTERING & CACHED VARIABLES
// ==========================================
float dist_history[3][5];
int history_idx[3] = {0, 0, 0};
float smoothed_distances[3] = {0, 0, 0};
float calibrated_distances[3] = {0, 0, 0};
bool first_reading[3] = {true, true, true};

volatile bool got_data = false;
float last_raw_val = 0;

uint8_t anchorBSSIDs[3][6];
int anchorChannels[3];
bool anchorFound[3] = {false, false, false};

float posX = 0.0;
float posY = 0.0;

WebServer server(80);

// ==========================================
// 3. THE WEB UI (HTML + JAVASCRIPT)
// ==========================================
const char index_html[] PROGMEM = R"rawliteral(
<!DOCTYPE html>
<html>
<head>
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>ESP32 Indoor GPS</title>
  <style>
    body { font-family: Arial; text-align: center; background-color: #222; color: #fff; margin: 0; padding: 20px;}
    canvas { background-color: #111; border: 2px solid #555; border-radius: 5px; box-shadow: 0px 0px 15px rgba(0,255,255,0.2); margin-top: 10px; max-width: 100%;}
    .stats { font-size: 1.2em; font-weight: bold; color: #00FFCC; margin-top: 15px; }
    .debug { font-size: 0.9em; color: #aaa; margin-top: 5px; }
  </style>
</head>
<body>
  <h2>Live Tracking (10m x 8m)</h2>
  <canvas id="mapCanvas" width="400" height="320"></canvas>
  <div class="stats" id="coords">X: 0.00 m | Y: 0.00 m</div>
  <div class="debug" id="dists">A1: 0m | A2: 0m | A3: 0m</div>

  <script>
    const canvas = document.getElementById('mapCanvas');
    const ctx = canvas.getContext('2d');
    const scale = 40;

    function drawGrid() {
      ctx.clearRect(0, 0, canvas.width, canvas.height);
      ctx.strokeStyle = "#333";
      for(let i=0; i<10; i++) {
        ctx.beginPath(); ctx.moveTo(i*scale, 0); ctx.lineTo(i*scale, 320); ctx.stroke();
        ctx.beginPath(); ctx.moveTo(0, i*scale); ctx.lineTo(400, i*scale); ctx.stroke();
      }
      ctx.fillStyle = "#00FFCC";
      ctx.fillRect(0, 320 - 10, 10, 10);
      ctx.fillRect(10 * scale - 10, 320 - 10, 10, 10);
      ctx.fillRect(5 * scale - 5, 320 - (8 * scale), 10, 10);

      ctx.fillStyle = "#fff"; ctx.font = "12px Arial";
      ctx.fillText("A1(0,0)", 15, 310); ctx.fillText("A2(10,0)", 340, 310); ctx.fillText("A3(5,8)", 215, 20);
    }

    function updateMap() {
      fetch('/data')
        .then(response => response.json())
        .then(data => {
          drawGrid();
          let cx = data.x * scale; let cy = 320 - (data.y * scale);
          if(cx < 0) cx = 5; if(cx > 400) cx = 395;
          if(cy < 0) cy = 5; if(cy > 320) cy = 315;

          ctx.beginPath(); ctx.arc(cx, cy, 8, 0, 2 * Math.PI);
          ctx.fillStyle = "#FF3366"; ctx.fill();
          ctx.strokeStyle = "#fff"; ctx.stroke();

          document.getElementById('coords').innerText = "X: " + data.x.toFixed(2) + " m | Y: " + data.y.toFixed(2) + " m";
          document.getElementById('dists').innerText = "A1: " + data.r1.toFixed(1) + "m | A2: " + data.r2.toFixed(1) + "m | A3: " + data.r3.toFixed(1) + "m";
        }).catch(err => console.log(err));
    }
    drawGrid(); setInterval(updateMap, 400);
  </script>
</body>
</html>
)rawliteral";

// ==========================================
// 4. FTM EVENT HANDLER
// ==========================================
void onFtmReport(arduino_event_t *event) {
  wifi_event_ftm_report_t *report = &event->event_info.wifi_ftm_report;
```

```
  if (report->status == FTM_STATUS_SUCCESS) {
    last_raw_val = (report->dist_est / 100.0); // Offset removed, handled entirely by LS Calibration now
    got_data = true;
  }
}

// ==========================================
// 5. SETUP
// ==========================================
void setup() {
  Serial.begin(115200);
  WiFi.mode(WIFI_AP_STA);
  WiFi.softAP("ESP32-Live-Map", "12345678");
  WiFi.onEvent(onFtmReport, ARDUINO_EVENT_WIFI_FTM_REPORT);
  WiFi.setSleep(false);

  server.on("/", []() { server.send(200, "text/html", index_html); });
  server.on("/data", []() {
    char json[100];
    sprintf(json, "{\"x\":%.2f, \"y\":%.2f, \"r1\":%.2f, \"r2\":%.2f, \"r3\":%.2f}",
            posX, posY, calibrated_distances[0], calibrated_distances[1], calibrated_distances[2]);
    server.send(200, "application/json", json);
  });
  server.begin();

  Serial.println("\n--- 10m Calibrated Tracker Starting ---");
  Serial.println("Scanning for Anchors...");
  int n = WiFi.scanNetworks(false, false, false, 150);
  for (int i = 0; i < n; i++) {
    for (int j = 0; j < 3; j++) {
      if (WiFi.SSID(i) == anchorNames[j]) {
        anchorChannels[j] = WiFi.channel(i);
        memcpy(anchorBSSIDs[j], WiFi.BSSID(i), 6);
        anchorFound[j] = true;
        Serial.printf("Found %s on Channel %d\n", anchorNames[j].c_str(), anchorChannels[j]);
      }
    }
  }
  WiFi.scanDelete();

  Serial.println("Connect Phone to Wi-Fi: ESP32-Live-Map");
  Serial.println("Open Browser to: http://192.168.4.1");
}

// ==========================================
// 6. MAIN LOOP
// ==========================================
void loop() {
  server.handleClient();

  for (int j = 0; j < 3; j++) {
    if (anchorFound[j]) {
      got_data = false;
      if (WiFi.initiateFTM(32, 0, anchorChannels[j], anchorBSSIDs[j])) {
        unsigned long start = millis();
        while(!got_data && millis() - start < 1000) { delay(10); server.handleClient(); }

        if(got_data && last_raw_val > 0.1 && last_raw_val < 30.0) {
          if (first_reading[j]) {
            for (int k = 0; k < 5; k++) dist_history[j][k] = last_raw_val;
            first_reading[j] = false;
          } else {
            dist_history[j][history_idx[j]] = last_raw_val;
            history_idx[j] = (history_idx[j] + 1) % 5;
          }

          float temp[5];
          for(int k=0; k<5; k++) temp[k] = dist_history[j][k];
          for(int p=0; p<4; p++) {
            for(int q=0; q<4-p; q++) {
              if(temp[q] > temp[q+1]) { float t = temp[q]; temp[q] = temp[q+1]; temp[q+1] = t; }
            }
          }
          smoothed_distances[j] = temp[2];
        }
      }
      unsigned long d_start = millis();
      while(millis() - d_start < 50) { server.handleClient(); delay(1); }
    }
  }

  if (!first_reading[0] && !first_reading[1] && !first_reading[2]) {
    calculateXY();
  }
}

// ==========================================
// 7. CALIBRATION & TRILATERATION
// ==========================================
void calculateXY() {
  // Apply Least Squares Calibration Math mapping measured distances to true distances
  calibrated_distances[0] = (smoothed_distances[0] * A1_m) + A1_c;
  calibrated_distances[1] = (smoothed_distances[1] * A2_m) + A2_c;
  calibrated_distances[2] = (smoothed_distances[2] * A3_m) + A3_c;

  float r1 = calibrated_distances[0];
  float r2 = calibrated_distances[1];
  float r3 = calibrated_distances[2];

  float d = anchors[1][0];
  float i = anchors[2][0];
  float j = anchors[2][1];

  float x = (pow(r1, 2) - pow(r2, 2) + pow(d, 2)) / (2 * d);
  float y = (pow(r1, 2) - pow(r3, 2) + pow(i, 2) + pow(j, 2) - (2 * i * x)) / (2 * j);

  if (isnan(x) || isnan(y)) return;

  posX = x;
  posY = y;
// Print raw vs calibrated data for debugging
  Serial.printf("RAW -> A1:%.2f A2:%.2f A3:%.2f | CALIB -> A1:%.2f A2:%.2f A3:%.2f | X:%.2f Y:%.2f \n",
                smoothed_distances[0], smoothed_distances[1], smoothed_distances[2],
                r1, r2, r3, x, y);
}
```

5.5 Results: ML System Performance

Overcoming Analytical Failures

- Pure mathematical trilateration fails in multipath-heavy indoor environments.
- By treating hardware biases as learnable patterns, the Ordinary Least Squares (OLS) model successfully mapped erroneous FTM data to functional true distances.

System Validation (Center-Grid Accuracy)

- Live Web UI tests demonstrate that the ML-calibrated system successfully stabilizes the tracking output.
- The tag's calculated position shows strong reliable accuracy when localized near the center of the 10m × 8m grid (e.g., test points $(6, 2)$ and $(4, 2)$).

5.6 Limitations: The Boundary Effect

Degradation Near Borders

- Visual outcomes reveal that positioning accuracy degrades significantly when the tag approaches the boundaries or corners of the grid (e.g., test points $(0, 0)$ and $(10, 0)$).

Root Cause: Non-Linear Multipath

- Linear Regression applies a *static, global* calibration weight to the distances.
- However, multipath reflections from physical walls create *highly non-linear* interference at the edges of the room, which a basic linear model cannot account for.

6 IMPLEMENTATION USING ESP32: RF FINGERPRINTING (KNN)

6.1 Overcoming Non-Linear Multipath

Because linear regression (OLS) applies a static, global calibration weight, it fundamentally fails at the physical boundaries of the room where multipath reflections are highly non-linear. To completely bypass geometric assumptions, the system architecture was transitioned to a purely data-driven pattern matching algorithm: K-Nearest Neighbors (KNN) RF Fingerprinting. Instead of fighting multipath delays, this approach utilizes the unique, messy signal reflections at any given coordinate as a distinctive "fingerprint" for that exact physical location.

6.2 Methodology and The "Thick Map"

The KNN implementation operates in two distinct phases to map and track the spatial environment:

- **Offline Phase (Calibration):** A "Thick Map" database was constructed by recording 50 raw FTM fingerprint sets across the 10m × 8m grid. Multiple datasets were combined for the exact same physical coordinates to capture natural environmental variance and drift, doubling the system's robustness.
- **Online Phase (Live Tracking):** The ESP32 continuously compares the live FTM distance readings (r_1, r_2, r_3) against the 50-point database in real-time by calculating the 3D Euclidean feature distance:

$$D = \sqrt{(r_1 - m_{r1})^2 + (r_2 - m_{r2})^2 + (r_3 - m_{r3})^2} \tag{7}$$

where m_{r1}, m_{r2}, m_{r3} represent the offline-mapped fingerprint distances at a specific index.

6.3 Inverse Distance Weighting (K=3)

While a strict Nearest Neighbor ($K = 1$) approach causes the tracking dot to rigidly snap between known grid points, a $K = 3$ configuration was deployed to enable smooth, continuous tracking. The algorithm identifies the 3 closest fingerprint matches in the database and averages their physical coordinates using Inverse Distance Weighting:

$$(X, Y)_{\text{final}} = \frac{\sum_{k=1}^{3} \left((X_k, Y_k) \cdot \frac{1}{D_k + \epsilon} \right)}{\sum_{k=1}^{3} \left(\frac{1}{D_k + \epsilon} \right)} \tag{8}$$

where ϵ is a small constant (e.g., 0.01) to prevent division by zero. This mathematical blending allows the ESP32 to accurately interpolate the tag's position even when it is located in the "dead zones" between the mapped calibration points.

6.4 Code Implemented

The following C++ snippet demonstrates the edge-computed KNN inference algorithm executed locally on the ESP32. This logic entirely replaces the previous geometric trilateration and OLS routines:

```
void calculateXY() {
  float r1 = smoothed_distances[0];
  float r2 = smoothed_distances[1];
  float r3 = smoothed_distances[2];
```

```

  // Arrays to hold the top 3 closest matches
  float best_dist[3] = {9999.0, 9999.0, 9999.0};
  int best_idx[3] = {-1, -1, -1};

  // 1. Search the 50-point database for the closest fingerprints
  for (int i = 0; i < NUM_POINTS; i++) {
    float d_r1 = r1 - mapR1[i];
    float d_r2 = r2 - mapR2[i];
    float d_r3 = r3 - mapR3[i];

    // Calculate Euclidean "Feature Distance"
    float sim_dist = sqrt((d_r1*d_r1) + (d_r2*d_r2) + (d_r3*d_r3));

    // Sort into the top 3 matches
    if (sim_dist < best_dist[0]) {
      best_dist[2] = best_dist[1]; best_idx[2] = best_idx[1];
      best_dist[1] = best_dist[0]; best_idx[1] = best_idx[0];
      best_dist[0] = sim_dist;     best_idx[0] = i;
    } else if (sim_dist < best_dist[1]) {
      best_dist[2] = best_dist[1]; best_idx[2] = best_idx[1];
      best_dist[1] = sim_dist;     best_idx[1] = i;
    } else if (sim_dist < best_dist[2]) {
      best_dist[2] = sim_dist;     best_idx[2] = i;
    }
  }

  // 2. Average the physical coordinates (Inverse Distance Weighting)
  float final_x = 0;
  float final_y = 0;
  float total_weight = 0;

  for (int k = 0; k < 3; k++) {
    if (best_idx[k] != -1) {
      float weight = 1.0 / (best_dist[k] + 0.01);
      final_x += mapX[best_idx[k]] * weight;
      final_y += mapY[best_idx[k]] * weight;
      total_weight += weight;
    }
  }

  // 3. Set the final interpolated coordinate
  if (total_weight > 0) {
    posX = final_x / total_weight;
    posY = final_y / total_weight;
  }
}
```

6.5 Results and Environmental Drift Management

Theoretically, the transition to a K-Nearest Neighbors (KNN) RF Fingerprinting approach was intended to neutralize the non-linear boundary errors observed in the OLS regression model by bypassing geometric assumptions entirely. However, empirical testing revealed that the KNN implementation ultimately yielded lower overall positioning accuracy compared to the previous OLS method. The primary limitation was the spatial sparsity of the calibration grid; the physical gaps between the mapped reference points caused the $K = 3$ Inverse Distance Weighting to inaccurately interpolate the tag's position in the unmapped spaces. While KNN successfully prevented the tracking coordinates from projecting outside the physical walls of the room, its average error within the center grid was higher than that of the generalized linear model.

Furthermore, the KNN model proved to be highly vulnerable to "Environmental Drift"—fluctuations in multipath signatures caused by moving furniture, open doors, or human presence over time. Because KNN relies on exact signal pattern matching, even minor dynamic changes in the room degraded the fingerprint matches, reducing accuracy further. To combat this constraint natively on the MCU, the system was upgraded with a **Manual Recalibration** feature via the asynchronous web interface. If the tag experiences localized drift, the user can manually input their known (X, Y) coordinate on the SoftAP website. The ESP32 immediately captures the new ambient FTM signature for that location, overwrites the specific index in the runtime memory, and utilizes the `Preferences` library to commit the updated map array directly to its non-volatile flash memory. While this allows the system to be maintained over time, the inherent environmental sensitivity of sparse RF fingerprinting made it practically less reliable than OLS regression for this specific hardware deployment.

7 Conclusion

This project systematically evaluated multiple approaches for developing a real-time Indoor Positioning System, transitioning from Software Defined Radios (ADALM-Pluto) to ESP32-based Wi-Fi Fine Time Measurement (FTM). Through rigorous empirical testing of geometric trilateration, Ordinary Least Squares (OLS) regression, and K-Nearest Neighbors (KNN) RF Fingerprinting, a definitive hardware limitation was established: achieving centimeter-level accuracy is fundamentally impossible using standard Wi-Fi FTM. Because the ESP32 operates on a narrow 20 MHz channel bandwidth, the physical radio pulses are too wide to distinguish the true Line-of-Sight (LOS) signal from overlapping multipath reflections in a cluttered indoor environment. Consequently, no numerical method, linear calibration, or machine learning algorithm can completely compensate for this physical-layer distortion. While ML techniques like OLS and KNN successfully stabilized the tracking output, prevented geometric boundary failures, and constrained errors to the sub-meter or meter level, the narrow transmission bandwidth remains the ultimate bottleneck for high-precision spatial localization.

8 Future Scope

To achieve the targeted centimeter-level accuracy, future iterations of this system must abandon narrowband Wi-Fi FTM in favor of true Ultra-Wideband (UWB) hardware modules, such as the Decawave DW1000 or DW3000 series transceivers. By utilizing a frequency bandwidth of 500 MHz or greater, dedicated UWB systems generate incredibly sharp, nanosecond-scale pulses. This extreme temporal resolution allows the receiving anchors to physically filter out delayed multipath wall reflections and lock exclusively onto the direct LOS signal, natively solving the indoor tracking problem without relying on heavy software calibrations or dense fingerprinting maps. Furthermore, integrating Sensor Fusion by adding an onboard Inertial Measurement Unit (IMU, such as the MPU6050) to the mobile tag would allow the system to cross-reference spatial coordinates with physical acceleration. This would effectively eliminate static environmental drift and dramatically smooth the real-time tracking trajectory between ranging polls.